

## CLT Q:

|    |                                                                                                                                                                                                                                                  |                                  |                                  |                                  |  |  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------|----------------------------------|--|--|
| 30 | <p>The half-life of <math>^{219}_{86}\text{Rn}</math> is 4.0 s.</p> <p>What is the activity after 3.4 s when the initial sample size at time <math>t = 0</math> s is 1.5 mol of <math>^{219}_{86}\text{Rn}</math>?</p>                           |                                  |                                  |                                  |  |  |
|    | <b>A</b> $1.4 \times 10^{19}$ Bq                                                                                                                                                                                                                 | <b>B</b> $8.7 \times 10^{22}$ Bq | <b>C</b> $1.3 \times 10^{23}$ Bq | <b>D</b> $7.7 \times 10^{23}$ Bq |  |  |
|    | <p><b>Answer: B</b></p> $A = A_0 e^{-\lambda t}$ $A_0 = \lambda N_0 = \frac{\ln 2}{4.0} (1.5 \times 6.02 \times 10^{23})$ $A = \frac{\ln 2}{4.0} (1.5 \times 6.02 \times 10^{23}) e^{-\frac{\ln 2}{4.0}(3.4)}$ $= 8.7 \times 10^{22} \text{ Bq}$ |                                  |                                  |                                  |  |  |

